



Multi-view syntax-semantics information bottleneck for dependency-driven relation extraction

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ABSTRACT

Dependency-driven relation extraction (DDRE) intends to improve the classification performance by exploring the syntax information contained in dependency tree. Although recent works have made impressive advances, they usually suffer severe challenges of losing sequential semantics information due to the pruning strategy. In addition, task-irrelevant information in representations such as noise or entity appositive information might be dominant in the procedure of representation learning, which interferes with the final performance of relation classification. To address these challenges, we propose a novel multi-view syntax-semantics information bottleneck (MS2IB) model, which aims at exploring supplementary information from a newly incorporated view and captures minimum sufficient information for the DDRE task. Specifically, MS2IB treats the DDRE task as a multi-view information learning procedure, where the sequential semantics information is supplemented under the guidance of representation learning with multiple views. Meanwhile, the minimum sufficient task-relevant information is extracted from the aspect of information compression and information preservation simultaneously. Finally, we formulate the objective of MS2IB as an information loss function based on the measurement of mutual information, where a new variational approach is presented to ensure its local optimum. Experiments on benchmark datasets and self-constructed dataset are conducted to show the superiority of MS2IB over the state-of-the-art models. For example, the proposed MS2IB achieves 91.42 % Precision, 89.21 % Recall and 90.28 % F1-score on the SemEval-2010 dataset. The MS2IB model further demonstrates strong generalization on domain datasets and robust performance on large-scale and noisy dataset. With the promising performance on both universal and domain datasets, the proposed model can be applied into various practical applications, such as information extraction and inference of transportation incidents/accidents.

1. Introduction

Unstructured text data has been one of the main data sources with the emergence of generative artificial intelligence. Text data includes a wealth of structured knowledge that can be applied into many applications, such as financial event relation extraction (Wan et al., 2023b), summary obfuscation detection (Taufiq et al., 2023) and autonomous vehicles (Zhang et al., 2022). Since extracting knowledge from text data is intractable and highly abstract, we usually resort to relation extraction,¹ i.e., dependency-driven relation extraction (DDRE) to learn the patterns in the unstructured text data (Qu et al., 2023). Recently, DDRE models have obtained impressive advancements in the domains of natural language processing and machine learning. DDRE aims to capture

syntax information by exploring and exploiting the dependency tree (Hong et al., 2020; Jin et al., 2022; Li et al., 2023a,b; Mandya et al., 2020; Song et al., 2019; Tian et al., 2021; Wan et al., 2023a; Yu et al., 2020; Zhang et al., 2018). However, it still confronts two significant challenges as Fig. 1 shows.

Ignoring the sequential semantics of pruned nodes. Most DDRE models perform pruning strategy from syntax perspective, leading to loss of sequential semantics information that is important to comprehend the context. For example, the pruned nodes “In” and “1835” of dependency tree, contain valuable sequential semantics information to identify the relation type *BornIn* between entity pair (Mark Twain, Florida). Intuitively, the nodes “In” and “1835” express Mark Twain’s birth year, while losing

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¹ Relation extraction aims at comprehensively understanding the context features to identify relation type between entity pairs of an input sentence.

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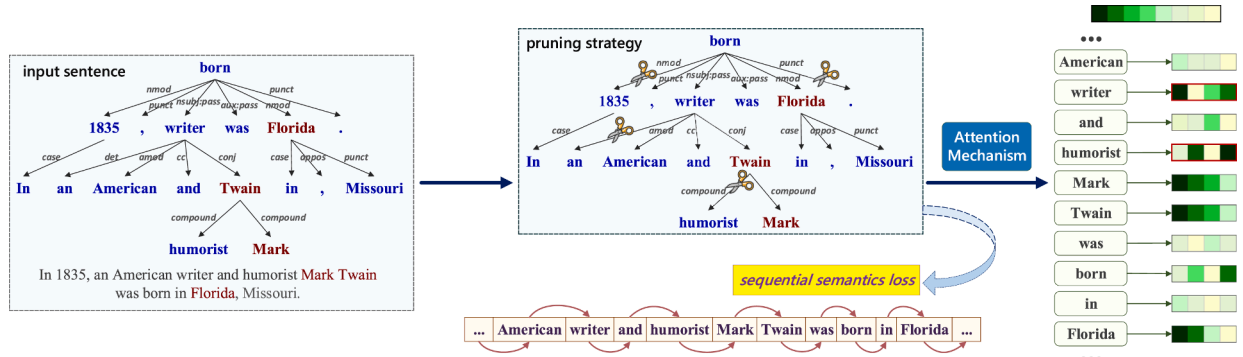


Fig. 1. Illustration of the challenging issues encountered by existing models. These DDRE models always focus on the pruning strategy and ignore the lost sequential semantics. Moreover, attention mechanism based DDRE models still suffer difficulties in capturing the task-relevant information for relation classification.

this *BornIn*-related information may degrade the performance of relation classification.

Ignoring the interference of task-irrelevant information. Almost all the DDRE models construct dependency tree by external tools that inevitably introduce irrelevant information, which leading to unsatisfactory performance of relation classification, especially confronting with datasets of low resources language in specific domain (e.g., Chinese railway dataset). Moreover, task-irrelevant information like entity appositive information may become dominant in the procedure of representation learning, thus potentially impairing the final results. For example, the writer and humorist are appositives of Mark Twain and contain outstanding *Person* features, while either of them associated with Florida as an entity pair can deliver the *BornIn* semantic relation. Obviously, the appositive information directly hampers the identification of relation between the actual entity pair (Mark Twain, Florida).

These two challenges are significant in the DDRE task. To tackle the aforementioned first challenge, one straight solution is to adopt graph neural networks to encode the dependency tree, aggregating information of neighbor nodes to learn localized patterns (Geng et al., 2022; Liang et al., 2023; Tian et al., 2021; Wu et al., 2024). Actually, these localized patterns mainly refer to task-relevant syntax information learned from a pruned dependency tree, failing to remedy the loss of sequential semantics information. For addressing the second challenge, a commonly employed strategy is to utilize an attention mechanism to diminish the importance of task-irrelevant information, thus boosting the performance of relation classification (Guo et al., 2019; Han et al., 2022). Unfortunately, this approach could be effective in addressing noise information, but it remains quite limited in dealing with the interference from task-irrelevant information such as entity appositives.

In this paper, we propose a novel multi-view syntax-semantics information bottleneck (MS2IB) model for DDRE task, which can automatically learn supplementary information from a newly incorporated view and explore the compression preservation mechanism of IB to obtain minimum sufficient task-relevant information. To facilitate the comprehension of the vital concepts of view and task-relevant information, here we define them as follows.

Definition 1. (View) View illustrates the data instance from different perspectives, such as part of speech, syntax and semantics, ensuring coherence across multiple viewpoints.

Definition 2. (Task-relevant information) Task-relevant information in DDRE task refers to significant features for identifying relation between entity pair. It is mainly the features between subject and object connected by the predicate in the syntax view, and the entity type features of sequential semantics in the semantics view.

Specifically, the MS2IB includes the following two modules: multi-view information learning and task-relevant information mining. For

the multi-view information learning, we devise view-specific encoders to learn representations of each view, where the syntax information is learned by the graph convolutional neural networks with an attention mechanism and the sequential semantics information is captured by the transformer mechanism. By this way, the learned sequential semantics can best serve as a remedy for the information loss of pruned nodes. For the task-relevant information mining, we explore the most relevant information from the aspects of information compression and information preservation simultaneously, in which the task-relevant information from multiple views can be obtained to learn DDRE-friendly representations. Additionally, we unify the two modules of MS2IB as an objective function measured by mutual information, which can perform the preservation of cohesive and relevant information across views and the compression of task-irrelevant information simultaneously. Finally, we design a novel variational optimization approach to ensure that the MS2IB achieves local optimum. Experimental results on benchmark and self-constructed datasets show the superiority of our MS2IB over the state-of-the-art baselines.

We summarize the main contributions of this work as follows:

- A novel MS2IB is proposed for DDRE task, which is capable of discovering supplementary information across views meanwhile mining task-relevant information through extending the IB theory.
- We devise a novel multi-view information learning approach to obtain sufficient representations across multiple heterogeneous views, where the sequential semantics of semantics view can be reasonably used to address the information loss in the syntax view.
- The task-relevant information mining is explored from the perspective of information compression and information preservation, where the minimum sufficient task-relevant features can be captured to learn DDRE friendly representations.
- We formulate a unified objective function of the MS2IB to optimize the multi-view information learning and task-relevant information mining simultaneously, where a dedicated variational optimization approach is implemented to ensure the attainment of a local optimum for MS2IB.

The rest of this paper is organized as follows. Section 2 simply analyzes the related studies about DDRE task and IB theory with deep neural networks. Section 3 we detail illustrate the proposed model. Experimental results are shown in Section 4. Finally, our conclusion and future work are given in Section 5.

2. Related work

2.1. Dependency-driven relation extraction

Many DDRE models obtain promising performance in various fields, such as structured knowledge acquisition (Jiang et al., 2024), web

information processing (Yang & Guo, 2022), low resource information extraction (Shi et al., 2021; Song et al., 2019; Yu et al., 2022) etc. According to the encoder strategy, we design a taxonomy of existing DDRE in the three categories: typical neural networks, attention-based networks, knowledge-enhanced networks. The typical neural networks aim to discover the dependency features by the LSTM (Geng et al., 2020; Jiang et al., 2024), graph convolutional neural networks (Hong et al., 2020; Jin et al., 2022; Li et al., 2023a; Mandya et al., 2020; Sun et al., 2020; Yu et al., 2020). The attention-based networks improve the performance of relation classification through alleviating the interference of noise features by attention mechanism (Guo et al., 2019; Luo et al., 2024; Tian et al., 2021; Xie et al., 2023; Zeng et al., 2024). The knowledge-enhanced networks usually introduce external knowledge (e.g., knowledge graph) to enhance representations to identify relation types (Gao et al., 2023; Hong et al., 2021). Although these DDRE models utilize dedicated neural networks to encode the dependency tree, they still fail to learn sequential semantics in the pruned nodes. It appears that introducing external knowledge has the potential to enhance the representations of input sentence, however, this strategy coupled with attention mechanism still difficult to learn the most meaningful task-relevant information.

2.2. Information bottleneck with deep neural networks

Recent years, the extraordinary success of deep neural networks has encouraged the progress of IB theory. The IB theory aims to compress the input variables X to a compact hidden variable Z , meanwhile maximally preserving the sufficient information about the output variable Y (Tishby et al., 1999). With the remarkable compression and preservation mechanism, it is used to analyze the deep neural networks from the perspective of information theory (Tian et al., 2024; Tishby & Zaslavsky, 2015; Wang et al., 2025, 2023). Now, the IB theory closely integrates with deep neural networks, which has been successfully applied in many applications, such as natural language edits (Marrese-Taylor et al., 2021), scientific document summarization (Ju et al., 2021), domain generalization (Li et al., 2022a), recommender systems (Wei et al., 2022) and multimodal named entity recognition (Cui et al., 2024). Recently, Wu et al. (2023) introduced the IB theory to the internal-information over-utilization and external-information exploitation issues in the multimodal relation extraction task. Though the IB theory achieves promising results on the above tasks, few researches concentrate on capturing the most relevant information for the dependency-driven relation extraction task.

3. The proposed model

3.1. Problem formulation

Let $S = \{x_1, x_2, \dots, x_{|S|}\}$ (x_i is a word) be an input sentence annotated with the entity pair (e_1, e_2) , where e_i means a continuous word sequence of the S . The relation extraction task is to predict the relation type $y \in Y$ between e_1 and e_2 , and the Y is a set of relation labels. There are two discrete variables S_1 and S_2 representing features captured from syntax view and semantics view. Afterwards, two joint distributions $\{p(Z, S_1), p(Z, S_2)\}$ can be constructed, where Z means minimum sufficient representations learned from the above multiple views. The proposed model aims to learn good representations $p(z|s_1, s_2)$ of the input sentence S with respect to the Y from the two relevant variables S_1 and S_2 . The details of our proposed MSIB are shown in Fig. 2.

3.2. Concept definition

The “syntax view” in the DDRE task is defined as follows:

Definition 3. (Syntax View) Let S_1 denote the syntax information that describes the dependency relation from the dependency tree of the input sentence S , such as subject-object relation between entity pair (e_1, e_2) .

To explore the syntax information, in this paper, we use the local and global dependency relation (Tian et al., 2021) about the mentioned entity of the input sentence, which is popular and successful in recent DDRE works. Specifically, we build standard graph convolutional networks (GCN) from dependency trees of the input sentence S and represent it as an adjacency matrix $\mathbf{X} = (x_{i,j})_{|S| \times |S|}$, where the $x_{i,j} = 1$ if there is a dependency relation between x_i and x_j , otherwise $x_{i,j} = 0$. The dependency types matrix $\mathbf{T} = (t_{i,j})_{|S| \times |S|}$ contains features dependency type (e.g., nsubj, conj, obj, etc.), capturing local dependency between words. Then we adopt the attention mechanism to capture the long dependency and GCN architecture to capture the overall syntax features as follows:

$$\begin{cases} \mathbf{X}_1 = ATT(Q = \mathbf{X}_1; K, V = \mathbf{X}_1) \\ S_1 = GCN(\mathbf{X}_1) \end{cases} \quad (1)$$

where $\mathbf{X}_1 = \mathbf{X} + \mathbf{T}$ is a vector concatenation operation between $x_{i,j} \in \mathbf{X}$ and $t_{i,j} \in \mathbf{T}$, $ATT(\cdot)$ means self-attention encoder and $GCN(\cdot)$ means standard GCN encoder. The details of syntax feature extraction are shown in Fig. 2(b).

Definition 4. (Semantics View) Let S_2 denote the semantics features that describes the sequential context semantics of the input sentence S , such as the type of an entity.

In contrast to the intuitive nature of image semantics, abstraction is the most remarkable characteristic and challenge of text semantics. For instance, *iron shoe* is a common railway term for wheel stopper, yet focusing solely on this phrase might lead one to interpret it as a shoe made of iron material. Primarily, this ambiguity in understanding arises from a lack of contextual coherence. Therefore, in this paper, we use Pre-trained language models to extract the general features of language framework, then introduce multi-head attention mechanism to learn context semantics of the input sentence S as follows:

$$\begin{cases} S_i = PreLM(S) \\ S_2 = MultiHead(Q = S_i; K, V = S_i)_{i=1}^{l=h} \end{cases} \quad (2)$$

where $PreLM(\cdot)$ represents encoder of Pre-trained language models, h means the number of attention head and $MultiHead(\cdot)$ means to concatenate h attention results.

3.3. The objective function

Obviously, the syntax information S_1 and semantics information S_2 respectively capture rich syntax features and semantics features for the DDRE task. Fusing information from these two views can effectively address the issue of sequential semantics loss resulting from pruning strategy. However, it still faces another issue that is interfered from task-irrelevant information (such as noise, appositives), impairing the final performance of DDRE task. The attention mechanism aims to refine representations by diminishing noise weight, while quantifying the extent of reduction remains challenging, moreover with lingering interference from appositives. As Fig. 2(c) shows, we propose a novel MS2IB to compress task-irrelevant information meanwhile preserve task-relevant information from syntax view and semantics view. We formulate the MS2IB as follows:

$$\mathcal{L} = \arg_{\max} \left[I(Z; Y) - \underbrace{\beta \left(I(S_1; Z) \oplus I(S_2; Z) \right)}_{\text{Information compression}} \right] \quad (3)$$

where Z means the minimum sufficient representations, β is a trade-off parameter to balance the information compression and information preservation, and $\oplus$ represents the union operation of two mutual information.

The proposed MS2IB offers three major merits over existing DDRE models:

- **View-wise supplementary information.** Unlike existing DDRE models utilizing encoders to remedy the loss of sequential semantics

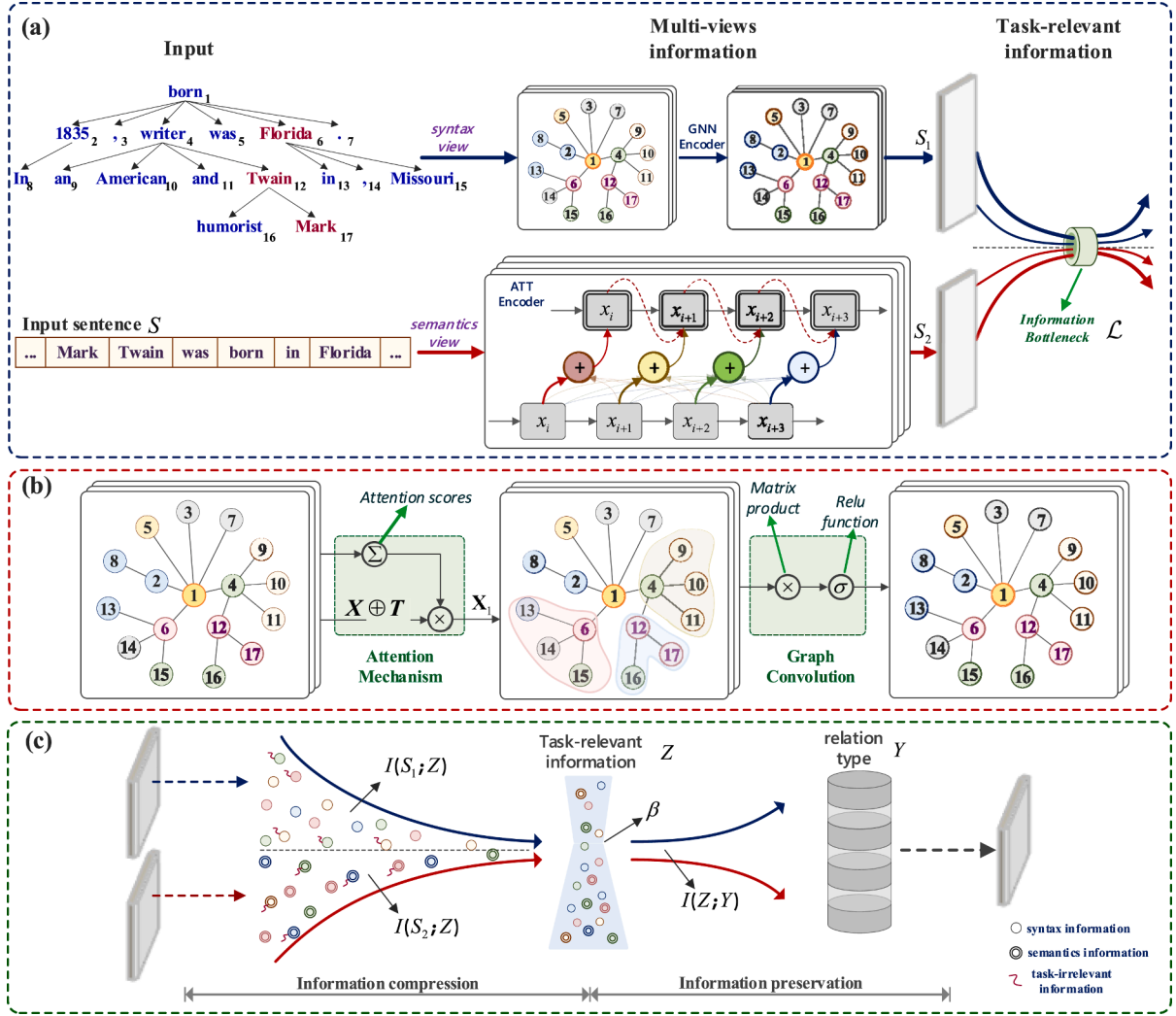


Fig. 2. (a) The overall architecture of the proposed MS2IB model. (b) The details of the syntax information learning. (c) The details of task-relevant information mining by IB theory. (Numerical labels in the graph nodes correspond to subscripted numbers in the dependency tree nodes).

information, MS2IB can fully exploit the supplementary semantics information through a dedicated semantics view.

- **Free from task-irrelevant information interference.** Instead of relying on attention mechanism to alleviate the interference of task-irrelevant information, a novel multiple views IB strategy is proposed to compress task-irrelevant information meanwhile preserve the task-relevant information.
- **Effective across diverse datasets.** The MS2IB is capable of handling datasets in various languages and diverse application fields, such as the RMSR dataset that describes unsafe production of railway work sections in Chinese that can be used for the identification of railway safe risk (see the Experiments section).

3.4. Optimization

To maximize the value of the objective function Eq. (3), we present a deep variational approximation to perform the optimization that benefits from the power of variational approach. It begins with mutual information $I(Z; Y)$ that measures the correlation between relation label Y and representations Z , where a larger value of $I(Z; Y)$ indicates more task-relevant features are captured in the representations Z . We present the details of $I(Z; Y)$ as

follows:

$$\begin{aligned}
 I(Z; Y) &= \int dy dz p(y, z) \log \frac{p(y, z)}{p(y)p(z)} \\
 &\geq \int dy dz p(y, z) \log r_1(y|z) + H(Y) \\
 &\geq \int ds dy dz p(s)p(y|s)p(z|s) \log r_1(y|z)
 \end{aligned} \quad (4)$$

where the $s \in S_1 \oplus S_2$ means information learned from both syntax view and semantics view, the $r_1(y|z)$ is a variational approximation with respect to $p(y|z)$, we use the Kullback Leibler divergence $KL[p(Y|Z), r_1(Y|Z)] \geq 0$ to obtain the first equality, the $H(Y)$ indicates the entropy about relation type Y , and the lower bound of $I(Z; Y)$ can be acquired by the Markov chain $Z \rightarrow \{S_1, S_2\} \rightarrow Y$.

The second term of Eq. (3) intends to compress task-irrelevant information, in which smaller values of mutual information $I(S_1, Z)$ and $I(S_2, Z)$ indicate less relevance between view-wise information and representations Z . To ensure the maximum of the objective function $\mathcal{L}$, we first introduce the theory of partial information function to serve as a fundamental basis for minimizing the total information learned from syntax and semantics views.

Lemma 1. For a collection of view sources $\mathbb{S}$, the partial information function f measures the information of partial view contributed uniquely through

each specific collection of $s \in \mathbb{S}$, it is given by

$$f(Z; s) = I_{\min}(\mathbb{S}; Z) - \sum_{t < s} f(Z; t) \quad (5)$$

where $I_{\min}(\mathbb{S}; Z) = \sum p(z) \min_{s_i} I(z; s_i)$ is redundancy information of all views that means the minimum information of each view provided for the representation Z , and the $t < s$ is an ordering relation in the redundancy lattice.

The above Lemma is used to decompose information and the remarkable trait of partial information function is non-negative, where detailed proofs for Lemma 1 can be seen in Williams and Beer (2010). Now, minimizing the total information from two views turns to discover upper bound of $I(S_1, Z) \oplus I(S_2, Z)$. We give the theorem as follows:

Theorem 1. Let $I(\{S_1, S_2\}; Z)$ be a multiple views interaction information, and $I(S_i; Z)$ be i th view information (or single view information) contributed to the representation Z . Then we have the following relationship:

$$I(S_1; Z) \oplus I(S_2; Z) \leq I(\{S_1, S_2\}; Z) \quad (6)$$

Proof. According to the Lemma 1, the interaction information can be decomposed as follows:

$$I(\{S_1, S_2\}; Z) = f(Z; S_1) + f(Z; S_2) + f(Z; \{S_1, \{S_2\}\}) + f(Z; \{S_1, S_2\}) \quad (7)$$

where $f(Z; S_1)$ and $f(Z; S_2)$ indicate the unique information that corresponding view contributes to the representation Z , $f(Z; \{S_1, \{S_2\}\})$ means the redundancy information between views and $f(Z; \{S_1, S_2\})$ represents additional information obtained through the combination of two views.

For the information compression term, we decompose it as follows:

$$\begin{aligned} I(S_1; Z) \oplus I(S_2; Z) &= (f(Z; S_1) + f(Z; \{S_1, \{S_2\}\})) \oplus (f(Z; S_2) + f(Z; \{S_1, \{S_2\}\})) \\ &= f(Z; S_1) + f(Z; S_2) + f(Z; \{S_1, \{S_2\}\}) \end{aligned} \quad (8)$$

Based on the non-negativity of partial information, we get an inequality $f(Z; \{S_1, S_2\}) \geq 0$ for the additional information. Then comparing the Eqs. (7) and (8), we directly get the inequality relationship as Eq. (6) expressed. The Fig. 3 facilitates a clearer understanding of the relationship between view information and interaction information with partial information decomposition. $\square$

According to the Theorem 1, the upper bound of information compression term can be formulated as follows:

$$\begin{aligned} I(S_1; Z) \oplus I(S_2; Z) &\leq I(\{S_1, S_2\}; Z) \\ &= I(\mathbb{S}; Z) \\ &\leq \int ds dz p(s) p(z|s) \log \frac{p(z|s)}{r_2(z)} \end{aligned} \quad (9)$$

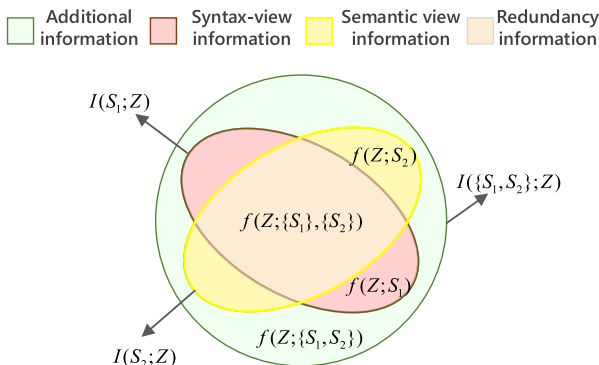


Fig. 3. Partial information decomposition for multi-view information learning.

where $r_2(z)$ is the variational approximation corresponding to the marginal distribution $p(z) = \int ds p(z|s)p(s)$ and we can obtain the upper bound of $I(S_1, Z) \oplus I(S_2; Z)$ by the inequality $\int dz p(z) \geq \int dz p(z)r_2(z)$ from the $KL[p(Z), r_2(Z)] \geq 0$.

Now, we integrate bounds of Eqs. (4) and (9) to obtain lower bound of the objective function $\mathcal{L}$ as follows:

$$\begin{aligned} \mathcal{L} &\geq \int ds dy dz p(s) p(y|s) p(z|s) \log r_1(y|z) - \beta \int ds dz p(s) p(z|s) \log \frac{p(z|s)}{r_2(z)} \\ &\approx \frac{1}{N} \sum_{n=1}^N \left[\int dz p(z|s_n) \log r_1(y_n|z) - \beta \left(p(z|s) \log \frac{p(z|s_n)}{r_2(z)} \right) \right] \end{aligned} \quad (10)$$

where the joint distribution $p(s, y) = p(s)p(y|s)$ is approximated via an empirical data distribution $p(s, y) = \frac{1}{N} \sum_n \delta_{s_n}(s) \delta_{y_n}(y)$, and N is the total number of sentence samples.

Maximizing $\mathcal{L}$ appears challenging, therefore, we reformulate this optimization problem as minimizing the negative value of $\mathcal{L}$. Suppose that the $p(z|s)$ obeys the Gaussian distribution $\mathcal{N}(\mu(s), \sigma(s))$, the reparameterization strategy is used to rewrite $p(z|s) dz = p(\epsilon) d\epsilon$, in which the $s = f(s, \epsilon)$ means deterministic function about s and Gaussian random variable ϵ . The objective of learning minimum sufficient information for DDRE task can be equivalently rewritten as follows:

$$\mathcal{L} = \frac{1}{N} \sum_{n=1}^N \mathbb{E}_{\epsilon \in p(\epsilon)} [-\log r_1(y_n | f(s_n, \epsilon))] + \beta KL[p(Z|s_n), r_2(Z)] \quad (11)$$

where the first term is the *CrossEntropy*($y, \hat{y}$) and $\hat{y}$ refers to the predictive label for the ground truth label y , and the second term means to maximally compress task-irrelevant information learned from two views.

We aim to minimize the above formulation and use stochastic gradient descent algorithm to implement, so that the minimum sufficient information from syntax and semantics views is captured. Algorithm 1 shows the details of proposed MS2IB model for DDRE task.

3.5. Complexity analysis

In this section, we present the complexity analysis of the MS2IB. In Algorithm 1, step 6 and step 8 involve the extraction syntax and semantics features with complexity of $O(\lambda_1 \cdot |S|^2 \cdot D \cdot B)$ and $O(\lambda_2 A \cdot D \cdot B)$, where λ_1 represents the layer number of GCN, $|S|^2$ means the size of dependency matrix T , D is the latent variable dimension initialized by pre-trained BERT model, B is the batch size, λ_2 is the number of Transformer layer and A is the attention head number. The complexity of steps 9 and 10 is $O(D \cdot B)$, which is governed by the variational approximation learning of latent variables Z . Therefore, the overall complexity of MS2IB model is $O(\lambda \cdot |S|^2 \cdot D \cdot B)$.

Algorithm 1 The MS2IB model for DDRE Task.

- 1: **Input:** Input sentence S with entity pair (e_1, e_2) , trade-off parameter β
- 2: **Output:** relation classifier
- 3: Initialize the representation of S with pre-trained BERT model in a distribution vector space, and obtain the dependency tree of S
- 4: **while** Multi-view features not convergence **do**
- 5: // extract syntax features
- 6: encode the S according to Eq. (1)
- 7: // extract semantics features
- 8: encode the S according to Eq. (2)
- 9: Train parameters according to the objective function Eq. (3)
- 10: adopt variational approximation to optimize parameters according to Eq. (11)
- 11: **end while**
- 12: **return** relation type

4. Experiments

In this section, our experiments are designed to address the following questions:

- **RQ1** How does MS2IB perform when compared with the SOTA baselines?
- **RQ2** Does the MS2IB model capture relevant features for final relation classification?
- **RQ3** How does the trade-off parameter β impact the performance of our proposed MS2IB?
- **RQ4** Can the MS2IB model generalize to other fields?
- **RQ5** How does the MS2IB model perform in terms of robustness on challenging datasets?
- **RQ6** Can the proposed MS2IB model effectively learn meaningful patterns?

4.1. Experimental settings

Datasets. In order to comprehensively evaluate the performance of our MS2IB model in DDRE task, we conduct extensive experiments on two types of datasets including universal domain and railway domain.

SemEval-2010 dataset. We adopt the classical dataset to evaluate the performance of our model, which is proposed by the SemEval-2010 Task 8 that focused on classification of semantic relations (Hendrickx et al., 2010). The SemEval-2010 dataset contains 10,717 samples including 8000 training samples and 2717 test samples. There are 10 relation types (including the “other” type), and the details of statistical information are shown in Table 1.

RMSR dataset. The RMSR (Railway Maintenance Safety Risk) dataset is constructed to validate the applicability of our proposed model, which is generated from the Railway Safety Monitoring System of the Chinese railway bureau. This dataset primarily documents safety inspectors’ records of unsafe practices in railway work sections. We dedicated over three months to annotating the RMSR dataset, with each sentence reviewed by three master volunteers. There are 23,414 sentences and 10,487 relation instances in this dataset. We utilize cross-validation to randomly select 80 % of the RMSR as the training set and the rest as the test set. There are four relation types and the details are demonstrated in Table 2.

Table 1
The statistical information of SemEval-2010 dataset.

Relation	Train	Test
Cause-Effect (CE)	1003	328
Instrument-Agency (IA)	504	156
Product-Producer (PP)	717	231
Content-Container (CC)	540	192
Entity-Origin (EO)	716	258
Entity-Destination (ED)	845	292
Component-Whole (CW)	941	312
Member-Collection (MC)	690	233
Message-Topic (MT)	634	261
Other	1410	454
total	8000	2717

Table 2
The statistical information of RMSR dataset.

Relation	Train	Test
R_{per}	6092	1524
R_{dev}	2839	710
R_{env}	53	14
R_{msg}	9745	2437
total	18,729	4685

Evaluation metrics. In this paper, we utilize the held-out evaluation to validate the performance of our proposed model. For these two datasets, we adopt the metrics of SemEval-2010 Task 8 that include the Precision (Prec.), Recall (Rec.) and F1-score (F1), which can compare with the strongest baseline more fairly. Note that larger metric values mean better performance of DDRE task.

Baselines. We compare the proposed MS2IB model with various state-of-the-art models, including non-dependency RE models, GNN-based DDRE models and ATT-based DDRE models. Specifically, non-dependency RE models perform the relation extraction task without incorporating syntax information from the dependency tree, GNN-based DDRE models leverage dedicated graph neural networks for dependency tree encoding and ATT-based DDRE models employ attention mechanism to mitigate the interference from noise information. We show the details as follows.

4.1.1. Non-dependency RE models

R-BERT: This model (Wu & He, 2019) focuses on exploring information from the pre-trained BERT model and target entities. It uses the architecture of pre-trained BERT to incorporate the encoding of entity pairs and obtains significant improvements on the SemEval-2010 dataset.

BERTEM + MTB: To improve the generalization ability of relation extraction models, Soares et al. (2019) proposed to extend Harris’ distributional hypothesis into relations and build representations of the task-agnostic relation from the entity-linked text. With the above advantages, the performance of BERTEM + MTB has been improved a lot.

BERT-DistMult: This model jointly trains knowledge representation learning model (i.e., DisMult) and pre-trained language model (i.e., BERT), which boosts representations of input sentence (Hong et al., 2021). The DisMult model provides more semantic connection information that aids context understanding, benefiting for RE task in the data-sparse application fields.

OSR: It introduces ontology concept to treat traditional RE task as named entity recognition, where the relation annotations are turned into ontology resource description framework (Bou et al., 2024). The OSR can be used for mining ontology documents, such as the Japanese in-house corpus.

4.1.2. GNN-based DDRE models

C-GCN: To capture more relevant information, Zhang et al. (2018) proposed improved graph convolutional neural networks, which use a novel pruning strategy on the input dependency tree. This model further improves the state-of-the-art performance.

C-GCN-MG: Mandya et al. (2020) proposed a multiple dependency sub-graphs based GCN architecture for relation extraction. This model can fully exploit advantages of the shortest dependency path and dependency graph of words linked to entities.

DP-GCN: This model (Yu et al., 2020) introduces a rethinking learning mechanism to prune the dependency tree instead of pre-defined rules. This dynamic pruning strategy helps to capture target-relevant features and remove those irrelevant features from the dependency tree.

ATERM-BERT: It splits the DDRE task into atomic relation encoding module and reasoning module, which aims to explore intermediate relations that contribute to final relation classification (Li et al., 2023a). The vanilla GNN is used to encode dependency tree and the LSTM is used for relation reasoning.

4.1.3. ATT-based DDRE models

C-AGGCN: The hard-pruning strategy of DDRE task may lose some valuable information in the dependency tree, a soft-pruning approach based on attention mechanism guided GNN is proposed by the C-AGGCN (Guo et al., 2019). It projects the overall dependency tree into a fully connected graph, where the weight of each edge is learned by a self-ATT mechanism.

Table 3
The settings of hyperparameters.

Hyperparameters	Values
GCN layer	2
transformer encoder head	8
transformer encoder num layers	4
learning rate	1e-5
warmup proportion	0.06
batch size	8
dropout rate	0.1

LST-AGGCN: This model (Sun et al., 2020) also concentrates on extracting relevant features from the dependency trees. Unlike the AGGCNs model, this model projects the dependency tree into a weighted graph, which fully considers the syntax dependency features and original structural dependency features.

A-GCN: This model (Tian et al., 2021) proposes an Attentive GCN architecture, which alleviates the effects of noise from the automatically generated dependency tree. The attention mechanism enables the model to obtain valuable dependency parser, selecting the important word dependencies as syntax information.

DPE: It intends to remove noise information through designing a dependency position encoder to incorporate dependency information and dependency types (Guo et al., 2022). The self-ATT mechanism is the core of DPE model, which assigns each dependency connection with specific weight.

BERT-GMAN: It extracts multi-granularity phrase information through constructing a key phrase extraction module based on the shortest dependency path (Xu et al., 2022). These key phrase features are further perceived by two attention mechanisms—Self-ATT mechanism and BiLSTM-ATT mechanism.

DAGCN: This model (Zhang et al., 2023) proposes a bidirectional information flow learning pattern, which enhances the representations of input sentence and improves the performance of DDRE task.

4.2. Hyperparameters settings

The proposed model is implemented in the PyTorch framework, running on a server equipped with Intel(R) Xeon(R) Silver 4216 CPU @ 2.10 GHz on 4 NVIDIA Tesla V100 GPUs (32 GB for each) and Ubuntu 14.04 operating system.² To better represent the input sentence, we introduce the notable pre-trained language model BERT to initialize context embeddings. Specifically, for the SemEval-2010 dataset, we use the BERT_{LARGE} model that includes 24 Transformer blocks with 16 self-attention heads, and the hidden size is 1024 following (Li et al., 2022b). Since the RMSR dataset is a newly constructed dataset without prior knowledge, context features are initialized via the popular model—BERT_{CHINESE} model, including 12 Transformer blocks with 12 self-attention heads and the hidden size is 768. We utilize the Standard CoreNLP Toolkits³ to construct a dependency tree for each input sentence. To further capture the features contained in the dependency tree, we follow (Tian et al., 2021) to encode the input sentence via a graph convolutional networks with an attention mechanism. We use the grid search to select the value of β from {1e-5, 1e-6, 1e-7, 1e-8, 1e-9, 1e-10, 1e-11}, and select the learning rate from {1e-5, 2e-5, 3e-5, 5e-5, 1e-4}. The more detailed parameters are shown in Table 3.

4.3. Performance analysis

Comparison with the SOTA (RQ1). In this part, extensive experiments are conducted to compare our MS2IB with many SOTA models and the

detailed results on the involved datasets are shown in Table 4. For a fair comparison, we directly report the results from the original papers on the SemEval-2010 dataset. As the RMSR is a newly constructed dataset, we tune hyperparameters to run all the compared models and report the best results. From the experimental results, there are some interesting observations and analyses showing as follows:

- **MS2IB VS. Non-dependency RE models.** MS2IB consistently outperforms all the compared Non-dependency RE models for the two datasets, especially on the specific domain dataset — RMSR. Obviously, Non-dependency RE models can be seen as single view learning models, which fully exploit semantics from context but ignore dependency information. As a remedy, BERT-DisMult model introduces DisMult module to significantly boost the performance of relation classification. It is not surprising since DisMult as a knowledge representation learning module can learn dependency relation between entity pairs. This phenomenon highlights the need for introducing the syntax information learning paradigm.
- **MS2IB VS. GNN-based DDRE models.** The proposed MS2IB obtains marvelous results compared to GNN-based DDRE models, which clearly demonstrates that jointly exploring the syntax information and semantics information across views play a significant role in prompting the DDRE performance. Actually, GNN-based DDRE models are inferior to our proposed multiple views learning approach, since only exploring syntax information can not guarantee a robust relation classification, particularly under low-resource language corpus. This emphasizes the need to integrate sequential semantics for enhancing the output of DDRE task.
- **MS2IB VS. ATT-based DDRE models.** ATT-based DDRE models are generally superior to the compared GNN-based models, since the attention mechanism alleviates the interference of noise information. However, these ATT-based DDRE models only assign different weights to each node that may not fully tackle intractable issue of entity appositive interference. With the effectiveness of IB theory for compressing task-irrelevant information and preserving task-relevant information, our proposed MS2IB learns minimum sufficient representations, obtaining better results than ATT-based DDRE models in terms of Prec., Rec. and F1 on the two datasets. In addition, this outcome also implies the applicability of our proposed model, which can be used for information extraction of intelligent railway transportation systems.

Effect of MS2IB (RQ2). An ablation study is given to further show the effectiveness of the core component in our proposed model. As shown in Section 3, our proposed model mainly includes a multiple views learning paradigm and IB strategy, which supplements sequential semantics features and captures task-relevant information for final relation classification. Therefore, specific view information and IB strategy are important in improving the performance of DDRE task. We then conduct ablation experiments on both SemEval-2010 and RMSR datasets to validate our MS2IB method and report the relation extraction results in Table 5.

As observed, the performance of relation classification on the SemEval-2010 dataset in terms of F1-score directly declines 1.39%, which indicates the importance of syntax information in learning dependency relationship between entity pair. The absence of semantics information supplement leads to a sharp decrease on both the SemEval-2010 dataset and RMSR dataset. For example, on the RMSR dataset, the Rec. and F1-score respectively decrease 7.1% and 4.86% compared with MS2IB. This large margin demonstrates that introducing semantics view information in effectively dealing with the sequential information loss caused by the pruned strategy. More importantly, we observe that a straightforward fusion of multi-view information may be counter-productive, for instance, the performance of MS2IB-IB is inferior to the MS2IB model, which lacks syntax view or semantics view. The above observations demonstrate the significant contribution of our proposed MS2IB to the final performance of relation extraction.

² The source code of the proposed MS2IB is provided at <https://github.com/SWT-AITeam/MS2IB>.

³ <https://stanfordnlp.github.io/CoreNLP/>.

Table 4

The performance of different models on SemEval-2010 and RMSR datasets. Bold values are SOTA results of corresponding metrics.

Model	SemEval-2010			RMSR		
	Prec.	Rec.	F1	Prec.	Rec.	F1
R-BERT (Wu & He, 2019)	–	–	89.25	77.29	77.28	77.15
BERTEM + MTB (Soares et al., 2019)	–	–	89.50	78.61	78.84	78.70
BERT-DistMult (Hong et al., 2021)	88.82	91.36	90.07	79.77	79.96	79.83
OSR (Bou et al., 2024)	–	–	89.90	79.82	78.46	78.43
C-GCN (Zhang et al., 2018)	–	–	84.80	70.27	61.08	65.35
C-GCN-MG (Mandya et al., 2020)	–	–	85.90	75.83	75.95	75.74
DP-GCN (Yu et al., 2020)	–	–	86.40	76.22	76.03	75.92
ATERM-BERT (Li et al., 2023a)	–	–	90.10	79.93	79.55	79.30
C-AGGCN (Guo et al., 2019)	–	–	85.40	76.81	76.13	76.04
LST-AGGCN (Sun et al., 2020)	84.43	87.56	86.00	77.36	76.85	76.72
A-GCN (Tian et al., 2021)	88.67	89.55	89.85	78.92	77.85	77.31
DPE (Guo et al., 2022)	–	–	90.2	79.65	79.89	79.74
BERT-GMAN (Xu et al., 2022)	89.64	90.97	90.25	80.00	80.17	80.06
DAGCN (Zhang et al., 2023)	–	–	86.00	76.52	75.53	75.38
MS2IB	91.42	89.21	90.28	80.30	80.39	80.29

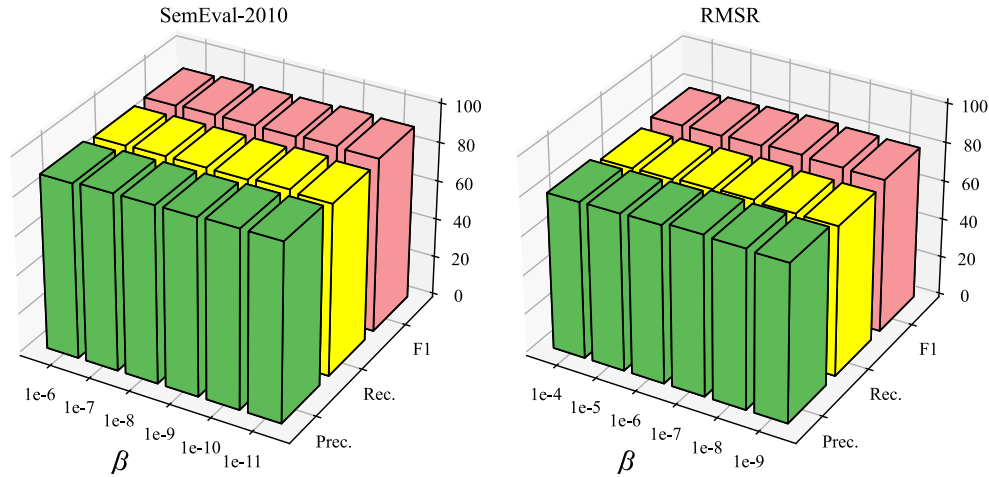


Fig. 4. Precision, Recall, F1 performance of MS2IB with different β values on the SemEval-2010 dataset and RMSR dataset.

Effect of different β values (RQ3). There is an important trade-off parameter β in the proposed model, helping MS2IB to capture task-relevant information to perform DDRE task. To delve deeper into the sensitivity of our proposed model with β , we conduct all the settings on the two datasets and present the results of relation classification as shown in the Fig. 4. Specifically, we adopt the grid search strategy to tune the optimal parameter β from the range $\{1e-4, 1e-5, 1e-6, 1e-7, 1e-8, 1e-9, 1e-10, 1e-11\}$. It is observed that all the metric values in each dataset tend to rise first and then decrease, which demonstrates that the MS2IB is slightly sensitive to the variations of β . This sensitivity empowers our proposed model to adeptly capture task-relevant information, particularly on dataset within the practical application fields, where the β well interprets the trade-off between compression information and preservation information.

Table 5

Ablation experiments on SemEval-2010 and RMSR datasets.

ID	Models	SemEval-2010			RMSR		
		Prec.	Rec.	F1	Prec.	Rec.	F1
1	MS2IB	91.42	89.21	90.28	80.30	80.39	80.29
2	- Syntax view	90.22	87.70	88.89	78.93	75.84	77.40
3	- Semantics view	88.67	89.55	89.03	77.68	73.21	75.43
4	- IB	89.76	88.20	88.77	76.66	74.84	74.23

Generalization analysis (RQ4). To evaluate the generalization capability of the MS2IB model across broader domains, we evaluate it on the GAD (Genetic Association Database) dataset (Bravo et al., 2015), which is a biomedical dataset archiving human gene-disease associations. As shown in Fig. 5, we compare the proposed MS2IB model to

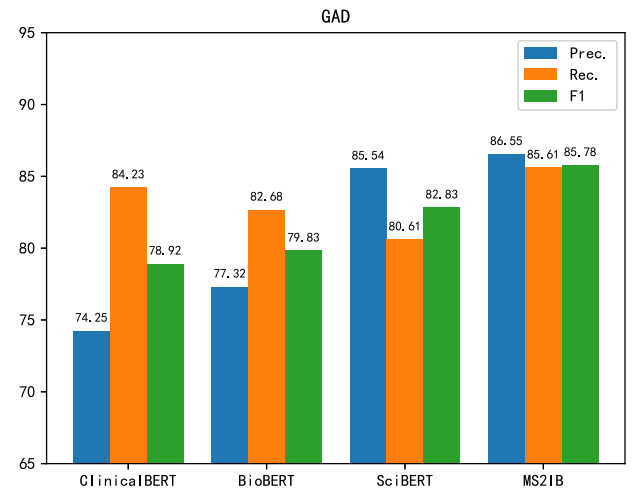
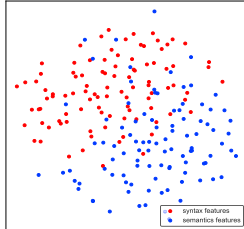
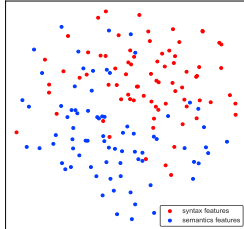
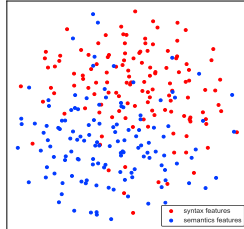
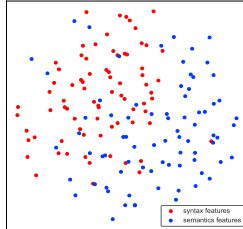


Fig. 5. Performance of MS2IB on the GAD dataset.

Table 6
Case study of the GAD dataset with t-SNE visualization.

Sample 1	Sample 2	Sample 3	Sample 4
No association between the -231 G > A polymorphism in the @GENE\$ gene and preeclampsia as well as any correlation with the main clinical features of the disorder were found, thus excluding a role for this polymorphism in susceptibility to @DISEASE\$.	The increased @DISEASE\$ rate of homozygous IL6-174G carriers was especially related to an increase in Gram-positive infections, and was not observed in VLBW infants who received prophylaxis with teicoplanin (frequency of Gram-positive sepsis in homozygous @GENE\$-174G carriers without prophylaxis 16.5 % versus 2.4 % in homozygous IL6-174G carriers with prophylaxis; $p = 0.033$).	We conclude that the tandem repeat polymorphism in the glycogen synthase gene and the polymorphisms at codons 513 and 972 of the @GENE\$ gene are not associated with a higher risk for the development of @DISEASE\$ in Japanese subjects.	Latex-@DISEASE\$ is associated with @GENE\$ *0201, DRB1 *0301, and *0901, as well as with HLA-DR functional group E, whereas latex-not-fruit allergy is associated with DQB1 *0202, and with both DRB1 *0701 and *1101 alleles.
			

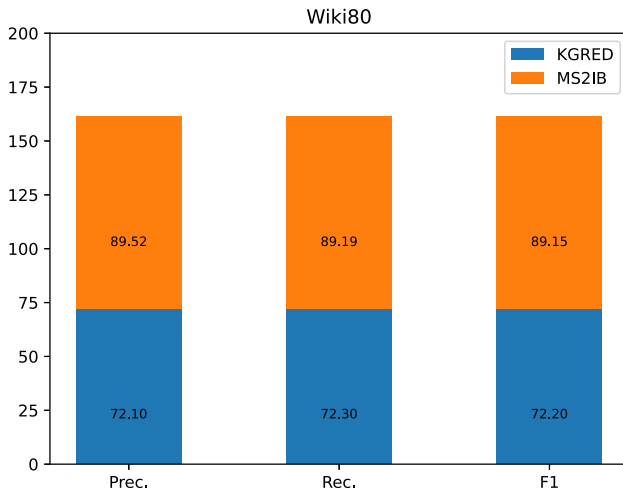


Fig. 6. Robustness analysis of MS2IB on the wiki80 dataset.

SOTA pre-trained language models—SciBERT, BioBERT and ClinicalBERT. Clearly, our MS2IB model outperforms all other methods across all metrics. It indicates that learning syntax and semantics features with IB strategy can learn compact and meaningful representations on the biomedical dataset. The good generalization ability of MS2IB method shows valuable potential in biomedical relation extraction.

Robustness analysis (RQ5). To validate the MS2IB model's effectiveness on large-scale and noisy dataset, we conduct an in-depth analysis using the Wiki80 dataset⁴. It is a challenging dataset that mainly used for weakly supervised relation extraction, since there is a semantic drift issue caused by mismatched sentence with rules (i.e., noise information) (Han et al., 2019). We compare the MS2IB model with the SOTA model—KGRED (Hou et al., 2024), enhancing contextual representations by leveraging knowledge graphs. The results are shown in Fig. 6. We observe that our MS2IB model outperforms the KGRED model, since

the effective compression–preservation mechanism of IB can well deal with task-irrelevant features or noise information.

Case study (RQ6). The primary strength of the proposed model lies in its ability to supplement lost sequential semantics across syntax view and semantics view, while concurrently capturing task-relevant features under the control of IB theory. To clearly illustrate the representation learning from syntax and semantic views on the challenging GAD dataset, we randomly selected four sentences and applied the t-SNE method to visualize the feature interaction between the two views. As Table 6 shows, the GAD dataset contains numerous biomedical terms that are difficult to understand for individuals without a background in biomedical knowledge. The MS2IB model enhances contextual understanding and achieves better relation classification by jointly leveraging syntax-semantic views representation learning. We further visualize the features with the heatmap as a case study to further validate the effectiveness of our proposed MS2IB. Specifically, we randomly pick four sentences from each dataset in SemEval-2010 and RMSR as Table 7. For each sentence, we compare features of proposed model as shown in Fig. 7 and Fig. 8. The top images show the dependency features captured by the syntax view, where predicate features are relatively clear, for example, “caused” and “called” of the first sentence in the first line of Fig. 7. The middle images show features integrated with sequential semantics, which enrich representations but also contain some interference information. For instance, the appositive “Toxocara canis” of entity “parasite” may interfere with relation extraction between entity pairs (Roundworms, parasite). Fortunately, we exploit the IB theory to learn minimum sufficient information by its core compression–preservation mechanism, and show the most relevant features in the bottom images. Obviously, the features of entity pairs are more outstanding than others, which confirms that our MS2IB can learn the most meaningful information to boost the performance of DDRE.

4.4. Discussion

DDRE task attracts intensive attention in the era of generative artificial intelligence, especially in the text mining with low resources language in specific application fields. It extracts vital syntax information from the dependency tree through the pruning strategy, thoroughly exploring the input sentence from the perspective of language structure without the need of external knowledge, such as knowledge

⁴ https://github.com/thunlp/OpenNRE/blob/master/benchmark/download_wiki80.sh

Table 7

Case Study on SemEval-2010 dataset and RMSR dataset. The entities in the triple are differentiated by colors here (e.g., the head entity with red color, the tail entity with blue color.). The first four sentences are from the SemEval-2010 dataset and the last four sentences are from the RMSR dataset.

Sentence	Triple
Roundworms or ascarids are caused by an intestinal parasite called Toxocara canis .	(Roundworms,Product-Producer,parasite)
The school master teaches the lesson with a stick.	(master,Instrument-Agency,stick)
The diseases are caused by gene mutations on the X chromosome.	(diseases,Cause-Effect,gene mutations)
The stories are narrated through dance.	(stories,other,dance)
跟班 检查 发现 , XX 站 9 道 4 # 轨 接头 3 块 大胶 溃烂 。	(轨 接头 , R_{dev} ,大胶 溃烂)
According to the inspection work, the 9 track No.4 rail joint of one station occurred that 3 large glues appeared aging and broken.	(large glues, R_{dev} ,aging and broken)
开 重点 任务 会 , 未 按 规 定 关 机 。 比 照 BGZ07006 给 予 考 核	(重点 任务 会, R_{mng} ,关机)
The machine was not turned off according to the management regulations, when the task-focused meeting was over.	(task-focused meeting, R_{mng} ,turned off)
排洪涵 淤积 造成 孔径 不足 (深度 超过 三分之一) 。	(排洪涵, R_{dev} ,淤积)
Discharge culverts occurred the circumstance of sedimentation that results an inadequate aperture (the actual depth of sedimentation exceed one-third of the diameter of the flood discharge culvert)	(Discharge culverts, R_{dev} ,sedimentation)
检查 XX 线路 维修工队 发现 班组 台帐 管理 严重 混乱	(维修工队, R_{per} ,台帐 管理 严重 混乱)
When inspected the maintenance team of one railway line, it was found that a serious circumstance – chaotic management of team ledgers.	(maintenance team, R_{per} ,chaotic management of team ledgers)

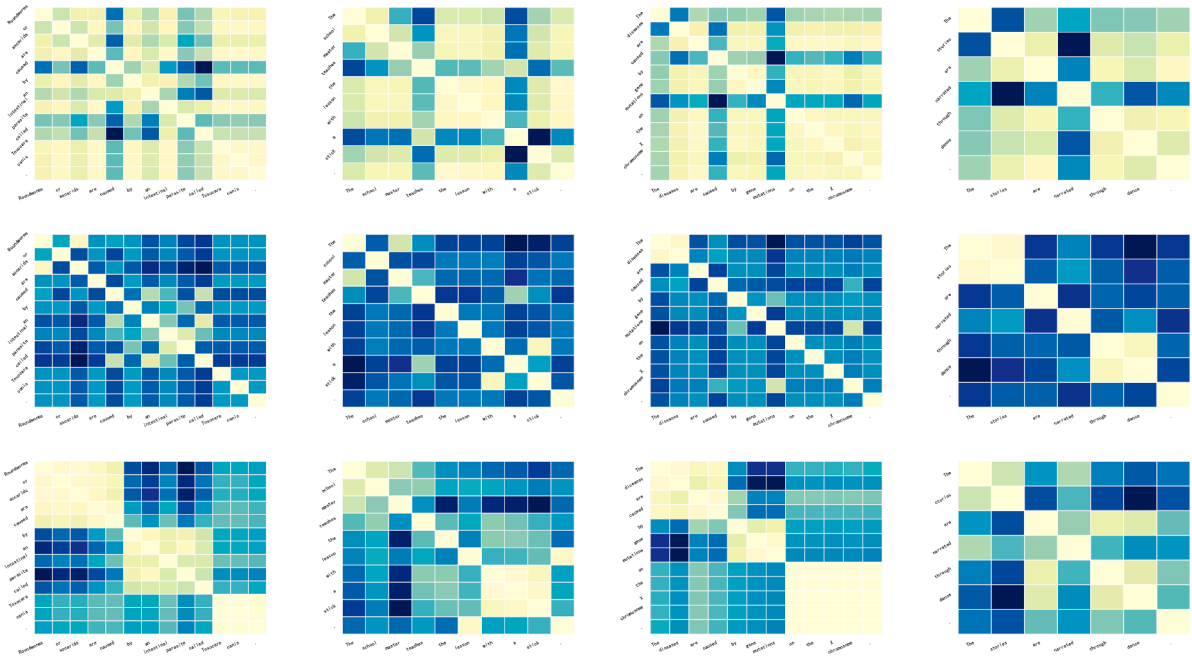


Fig. 7. Features comparison on SemEval-2010 dataset. The top, middle and bottom images respectively show heatmaps of dependency features, multi-views features and task-relevant features.

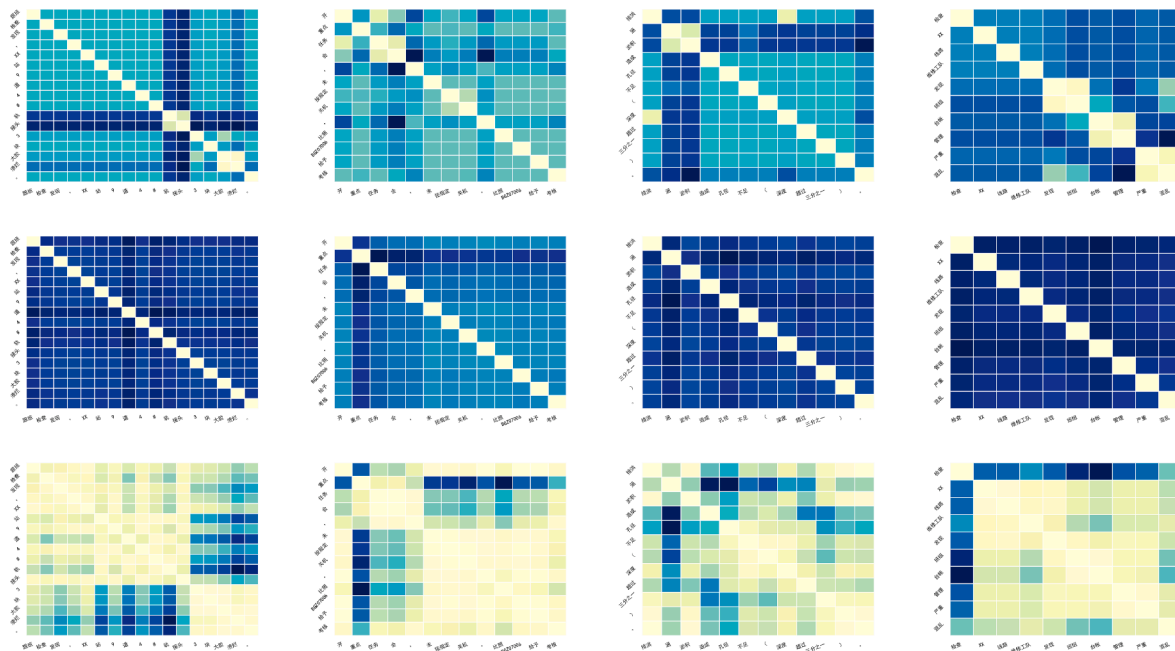


Fig. 8. Features comparison on RMSR dataset. The top, middle and bottom images respectively show heatmaps of dependency features, multi-views features and task-relevant features.

base. However, the lost sequential semantics issue of pruned nodes and interference of task-irrelevant information issues still limits the performance of DDRE task. Actually, existing DDRE models focus on designing dedicated deep neural networks (Hu et al., 2024; Wen et al., 2020) to learn sufficient information, which can be seen as a single view learning paradigm based on the syntax view. Our proposed MS2IB model learns sufficient information from syntax view and semantics view, meanwhile mines task-relevant information by a novel IB method. Experiments on both benchmark dataset and domain dataset have shown the superiority of our proposed model. Especially for the domain dataset—RMSR, it thoroughly demonstrates the valuable practical application in the specific domain text mining. For example, the MS2IB model can be used to construct specific domain knowledge graph, more importantly, it provides meaningful and interconnected knowledge (i.e., triple knowledge) for specific domain inference tasks, like predicting railway safety risk and analyzing railway safety accidents.

5. Conclusion

This paper investigates how to supplement lost sequential semantics and mine task-relevant information in the scenario of DDRE. Our MS2IB model takes advantages of both the syntax view and semantics view to capture sufficient information for representations, meanwhile learning minimum task-relevant information to boost the overall performance of relation classification. From theoretical and empirical perspectives, it demonstrates the effectiveness of the proposed optimization approach for our MS2IB model. Experimental results on both the English benchmark dataset and the Chinese self-constructed dataset have shown the superiority of the proposed MS2IB model in comparison to the state-of-the-art models. However, the different external tools for constructing dependency trees always generate inconsistent dependencies, it would be difficult to address this inconsistent problem and thus the performance of relation classification may be affected. In addition, since the proposed MS2IB runs on a universal pre-trained language model, it is sometimes insufficient in initializing the context representations when confronted with specific domain text in low resource language, such as Chinese railway text. Finally, the

multi-modal features are also important for relation extraction, which is worth considering to inspire some ideas from the multi-modal based NLP models.

In the future, we try to solve the above limitations, especially the challenging problem of insufficient representations. Additionally, it may be valuable to extend our model into more practical scenes, such as the identification of railway safety risks and incidents/accidents inference.

CRedit authorship contribution statement

Wei She: Conceptualization, Methodology, Software, Validation, Supervision; **Xi Wang Li:** Methodology, Formal analysis, Software, Writing - review & editing; **Linpu Lv:** Data curation, Software, Investigation; **Youwei Wang:** Conceptualization, Methodology, Validation, Writing - review & editing; **Honghui Dong:** Conceptualization, Supervision; **Zhao Tian:** Methodology, Validation, Supervision.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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